

Fig. 3: An actual-size, single-side PCB for the line-follower robot

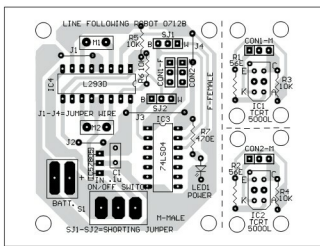


Fig. 4: Component layout for the PCB of Fig. 3

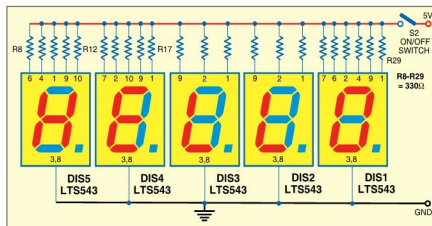


Fig. 5: Circuit for HELLO

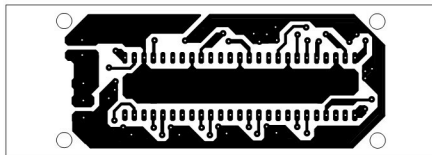


Fig. 6: An actual-size, single-side PCB for Fig. 5

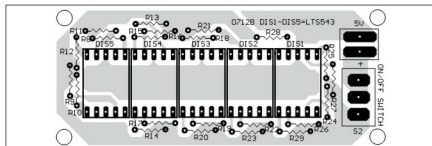


Fig. 7: Component layout for Fig. 6

turns left. When the right sensor, say, IC1, detects the black track, the robot turns right. When both the sensors are in white surface then, the robot moves in forward direction.

If you want the robot to follow a white track change the jumpers' positions towards W terminals. As the reflector sensor moves on the black surface, the emitted signals are absorbed by the black surface and both the inverters (N1 and N2) invert to low. These low signals are fed to the motor driver L293D and the robot moves forward. When the left sensor, say IC2, detects the white track, the robot turns left. When right sensor, say IC1, detects the white track, the robot turns right. When both the sensors are in black surface then the robot moves in forward direction.

The circuit is powered by a 12V DC battery. IC 7805 is used to provide regulated 5V supply to the circuit. Capacitor C1 bypasses the ripples from the regulated supply. LED1 acts as the power-on indicator and resistor R7 limits the current through LED1.

Construction and testing

An actual-size, single-side PCB for the line-follower robot is shown in Fig. 3 and its component layout in Fig. 4.

The sensors sections should be separated from the main PCB by cutting along the dotted lines. Right and left sensors should be mounted on